

Required Capstone Checkpoint 1.1: Evaluating When Retrieval Is Required

Directions

Select one of the provided capstone scenarios and use it as the foundation for this checkpoint. You will continue developing this same scenario throughout the program. Download this worksheet and save a copy to your device before you begin working. Type your responses directly into the document, save your completed work, and upload the finished file when you are ready to submit.

Use this worksheet to:

- Analyze how an LLM performs without retrieval
- Document prompts and corresponding model outputs
- Identify limitations and patterns in model behavior
- Determine when retrieval is required
- Reflect on how retrieval quality affects reliability and user trust

1. Selected Scenario and Intended Use

Select your capstone scenario:

- ☒ Research Paper Navigator
- ☐ Wikipedia Retrieval Engine

Briefly describe the kinds of questions, tasks, or user interactions your system should support.

The system is a conversational interface over a collection of 153 research papers spanning roughly 2009–2024. Intended users are researchers and engineers who need to locate and verify specific claims in this literature without reading every paper. It should support: **Single-paper questions** — a specific finding, method, or result from one named paper, returned with a direct quotation from that paper. **Cross-paper comparison** — how two or more papers define a problem, differ in approach, or report conflicting results. **Trend questions over time** — how an idea evolved across the collection, which requires reasoning over publication dates and not only content. **Grounded responses** — every substantive claim traceable to a named source and quoted verbatim. **Multi-turn conversation** —

follow-up questions that refer back to papers already discussed. **Corpus-boundary honesty** — stating plainly when an answer is not in this collection, rather than answering from general knowledge.

2. Prompt and Output Evidence

- Provide two prompts relevant to your scenario. Where possible, use prompts that test different types of information needs or tasks.
- Include the corresponding LLM outputs for each prompt.

Prompt 1 and Corresponding LLM Outputs

PROMPT: In the paper 'An empirical study of adaptive concretization for parallel program synthesis', what conclusions are made? Quote the abstract directly.

RESPONSE: I'm not able to quote the abstract directly because I don't have the paper text available in this chat.

Prompt 2 and Corresponding LLM Outputs

PROMPT: Which papers in this collection address pacmpl, and document and compare the progress made in them? Cite each paper.

RESPONSE: I'm not sure which collection you mean, and I don't have the papers themselves here to inspect.

If you share the collection name, a link, or upload the papers/PDFs, I can identify which ones address **PACMPL** and then **document and compare the progress** made in each, citing each paper directly.

If helpful, I can also help you search the collection by looking for:

- titles mentioning **PACMPL**
- references to the **Proceedings of the ACM on Programming Languages**
- papers that cite or discuss PACMPL venues such as **OOPSLA**, **POPL**, **PLDI**, etc.

Send the collection and I'll do the comparison.



3. Analysis of Model Limitations

- Identify key limitations in the model's responses, such as hallucinations, knowledge gaps, stale information, inconsistency, vagueness, or missing information.
- Use specific prompt/output examples to support your analysis.

Key limitations and supporting prompt/output examples

Response: Because the model cannot access the underlying paper documents, it lacks sufficient context to fulfill most user requests. Furthermore, as demonstrated in Prompt 1, the model generated a fabricated quote that does not exist within the authentic publication text.

4. Patterns in Model Behavior

- Describe any meaningful patterns or trends you observe across the outputs.
- Consider whether certain types of prompts produce more reliable responses than others.

Meaningful patterns or trends

Response: When lacking source access, the model prompts the user to supply the missing documents or collection details. Conversely, when asked for exact textual citations, it generates plausible-sounding fabricated passages rather than admitting its inability to locate the content, thereby damaging user confidence in its outputs. In the case of Prompt 1, I guess that it used the extra information given in the prompt about the paper “(GU et al. ACL 2024)” to find out some information from the internet and quote some hallucinated response and it did not mention that it found it out from the wild.

5. Determining Whether Retrieval Is Required

- Explain whether retrieval is required for your scenario and identify which observed limitations retrieval could help address.
- Justify your reasoning using evidence from the outputs.

Explanation and justification

Response: Response: Retrieval mechanisms are essential for these tasks since the underlying neural system lacks direct indexing or visibility into the actual document text. Lacking document grounding, the interface fails to deliver verifiable verbatim passages or pinpoint exact findings, resulting in evasive or fabricated outputs. Incorporating a dedicated retrieval infrastructure empowers the system to pull relevant text directly, ensuring responses remain grounded and strictly faithful to the underlying literature collection.

6. Retrieval Requirements

- Describe the types of information that would need to be retrieved to improve reliability, grounding, and response quality. Consider whether the information is proprietary, frequently changing, highly specific, or otherwise unavailable to the model without retrieval.
- Explain when retrieval would or would not be necessary.

Information that would need to be retrieved

Response: Any information that the model is not trained on, it needs a source of truth, which may not be available to it due to some reasons like being offline, or the information not available publicly or it changes too fast for a model to be trained on. Such sources of information need to be retrieved from a dedicated and provided corpus.

When retrieval would or would not be necessary

Response: To deliver grounded, faithful answers, the infrastructure must pull the specific titles and abstracts corresponding to target papers. Beyond metadata, accessing complete document body texts is essential for retrieving verbatim citations, contrasting distinct problem formulations across publications, and determining which studies tackle PACMPL themes. Without indexing these detailed primary sources, the system cannot reliably evaluate paper conclusions, contrast methodologies, or fulfill requests for exact textual evidence.



7. Reflection and Next Steps

Explain how the precision, relevance, and accuracy of retrieved information affect output quality, system reliability, and user trust. Consider how retrieval quality might improve or degrade system performance. What could happen if the retrieved information is incomplete, loosely related, outdated, or inaccurate?

Explanation

Response: The precision, accuracy, and relevance of retrieved context directly dictate output quality and system dependability. When a pipeline yields tangential or imprecise passages, the model risks propagating fabrications, misrepresenting scientific insights, and damaging user trust. Conversely, high-quality retrieval guarantees that every generated response remains firmly grounded—enabling exact textual quotations, rigorous cross-paper comparisons, and faithful syntheses that reinforce user confidence in the platform.

8. Retrieval Requirements

- Reflect on what you learned about LLM limitations, common failure modes, and retrieval needs from this activity.
- Describe what you would investigate or improve next as you continue developing your capstone system.

Reflection

Response: LLMs are limited in helping beyond what they are trained without access to any new information. Sometimes, they hedge their answers and other times they confidently give incorrect answers. It is critical to prove reliability of LLMs before exposing them to users to avoid mistrust.

I want to discern an unreliable answer with the hallucinated citations vs accurate ones. Probably the LLM can publish a confidence score or run another agent to critique its answer.

Supporting Materials

Use this section to list, describe, or link to any supporting materials included with your submission.

Supporting material	Location, link, or note
I used Claude code to help me draft some answers and give me feedback on my original answers.	Add notes, links, or locations here

End of Required Capstone Checkpoint 1.1

Return to the assignment page in the learning platform to upload and submit your completed capstone checkpoint and complete any remaining items in this module.